

NAME: NEPTUN CODE:

Probability Theory exam, January 10th, 2023

Working time: 100 min. Only simple, non-programmable calculators are allowed.

Maximum score: 110 points, but we consider 100 points already as 100%.

The. 1. Let X be a Poisson random variable with parameter $\lambda > 0$.

- (a) (6 points) Define the probability weight function of X . Prove the formulas for the expected value and variance of X .
- (b) (8 points) State and prove the Poisson approximation theorem of binomial random variables.
- (c) (5 points) We have a box with a lot of lightblubs in it; each of which is faulty with a small probability. We know that the probability of having zero faulty lightblubs is $2/3$. What is the probability that there are at most 2 faulty blubs in the box?

The. 2. Consider two random variables X, Y .

- (a) (4 points) Define their covariance and prove that $\text{Cov}(X, Y)$ is bilinear. Name the properties of expectation when you use them.
- (b) (2 points) Prove that $\text{Cov}(X, Y) = 0$ if X, Y are independent. Name the properties of expectation when you use them.
- (c) (7 points) Is it also true, that X, Y are independent if $\text{Cov}(X, Y) = 0$? Why?

The. 3. Let $\lambda, \mu > 0$ and let $X \sim \text{Poi}(\lambda), Y \sim \text{Poi}(\mu)$ be independent random variables.

- (a) (4 points) Calculate the moment generating function of X .
- (b) (4 points) Using moment generating functions, prove that $X + Y \sim \text{Poi}(\lambda + \mu)$. Explain all steps of your proof.
- (c) (10 points) Let $Z|X \sim \text{Bin}(X, p)$ for some constant $p \in [0, 1]$. Prove that $Z \sim \text{Poi}(p\lambda)$.

Prac. 1. (20 points) To celebrate Advent, Anne and Belle made tea selections for each other. Both selections consist of 24 teabags; placed onto one another in a small gift box. The two girls organized the teabags randomly, choosing each permutation with the same probability, independently of each other. There are exactly 8 black and 6 green teas in a gift box. On each day of Advent, they drink the uppermost tea in their respective box, and they only drink one tea per day.

Let X denote the number of days when they both drink black tea. Calculate $\mathbb{E}[X]$ and $\text{Var}(X)$.

BONUS: (10 points) Let Y denote the number of days when they both drink green tea. Calculate $\text{Cov}(X, Y)$.

Prac. 2. (15 points) First, we roll with a fair die and write X for the result. Then, we keep throwing two fair dice until the sum of the rolled numbers satisfy a certain condition depending on X . If X is smaller than 5, then we stop when the sum is even. Otherwise, we stop when the sum is divisible by 3. What is the probability that the sum is bigger or equal to 10 when we stop?

Prac. 3. Jack arrives to school with a delay of $X \sim \text{Uni}[0, 10]$ minutes. Jill arrives with a delay of $Y \sim \text{Uni}[0, 15]$ minutes, independently of Jack.

- (a) (5 points) What is the probability that Jill arrives earlier than Jack?
- (b) (10 points) What is the expectation of the difference $|X - Y|$?

NAME: NEPTUN-CODE:

Probability theory 1 exam, 2023. Dec. 19.

*Working time: 100 min. Only simple, non-programmable calculators are allowed,
standard normal distribution table on the other side.*

The achievable maximum score (with the Bonus exercise) is 110 points, but we consider 100 points as 100%.

- T. 1.** (a) (2+4 points) Define the notion of variance and prove the formula about the value of $\text{Var}(aX + b)$ (where $a, b \in \mathbb{R}$ and X is a random variable).
- (b) (2+4 points) State a necessary and sufficient condition for $\text{Var}(X + Y) = \text{Var}(X) + \text{Var}(Y)$ to hold using the notion of covariance! Prove this statement.
- (c) (3+3 points) Calculate the expected value and the variance of a random variable X with binomial distribution by writing X as a sum of indicator variables.
- T. 2.** Let X denote a random variable with $\text{EXP}(\lambda)$ distribution.
- (a) (2+6 points) Write down the probability density function of X and calculate the moment generating function of X .
- (b) (5 points) Let $\psi(x) = e^{-\lambda x}$. What is the distribution of $\psi(X)$?
- (c) (3+2 points) State and prove the memoryless property of the $\text{EXP}(\lambda)$ distribution.
- T. 3.** (a) (4 points) State and prove Markov's inequality!
- (b) (5 points) State and prove Chebyshev's inequality!
- (c) (5 points) State and prove the weak law of large numbers!

Formula sheet: $X \sim \text{EXP}(\lambda)$: $\mathbb{E}(X) = \frac{1}{\lambda}$, $\text{Var}(X) = \frac{1}{\lambda^2}$. $X \sim \text{GEO}(p)$: $\mathbb{E}(X) = \frac{1}{p}$, $\text{Var}(X) = \frac{1-p}{p^2}$.
 $X \sim \text{POI}(\lambda)$: $\mathbb{E}(X) = \text{Var}(X) = \lambda$.

- P. 1.** In a town two taxi companies operate, one uses yellow cabs, the other white cabs. 10% of the taxis are white, the rest of them are yellow. A traffic accident caused by a taxi driver was witnessed by a bystander, who claims that the culprit was driving a white taxi. Subsequent police investigation found that the reliability of the witness is 80%, meaning that under similar circumstances the witness can correctly identify the colour of a taxi in 80% of the cases.
- (a) (8 points) Given the above info, what is the probability that the culprit was indeed driving a white taxi?
- (b) (8 points) Another independent witness (also with 80% reliability) shows up and confirms the claim of the first witness. Now what is the probability that the accident was caused by a white taxi?
- P. 2.** (16 points) The sides of a rectangle have length 1 and 2. On both of the sides of length 1 we pick one point each, independently, with uniform distribution. Let Z denote the distance of these two points. Determine the cumulative distribution function of Z .
- P. 3.** An ice cream stand sells two types of ice cream: chocolate and vanilla. The customers arrive according to a Poisson point process, where the expected time between two consecutive customers is two minutes. Each customer chooses between the two types of ice cream with a fair coin toss.
- (a) (7 points) Identify the cumulative distribution function of the arrival time of the first customer who buys chocolate after the ice cream stand opens.
- (b) (11 points) Use the central limit theorem to estimate the probability that the number of customers that arrive between 8am and 2pm differs from the number of customers that arrive between 2pm and 8pm by at most ten.

Bonus: (10 pont) Estimate the probability that between 8am and 2pm more people buy vanilla than chocolate, but between 8am and 8pm more people buy chocolate than vanilla.

NAME: NEPTUN-CODE:

Probability theory 1 exam, 2024. Jan. 9.

*Working time: 100 min. Only simple, non-programmable calculators are allowed,
standard normal distribution table on the other side.*

The achievable maximum score (with the Bonus exercise) is 110 points, but we consider 100 points as 100%.

- T. 1.** (a) (5 points) Define when the events $A_1, \dots, A_n$ are mutually independent.
- (b) (6 points) Construct events A, B, C such that $\mathbb{P}(A) = \mathbb{P}(B) = \mathbb{P}(C) = 1/2$, moreover A, B, C are pairwise independent, but not mutually independent.
- (c) (5 points) Let us assume $\mathbb{P}(B) > 0$. Show that A and B are independent if and only if $\mathbb{P}(A) = \mathbb{P}(A|B)$.
- T. 2.** Assume that the random variable X is absolutely continuous, let us denote its cumulative distribution function (c.d.f.) by F and its probability density function (p.d.f.) by f .
- (a) (8 points) Let $\psi : \mathbb{R} \rightarrow \mathbb{R}$ be a differentiable, strictly monotone bijection. Let $Y = \psi(X)$. Write down and prove the formula that expresses the c.d.f. G and the p.d.f. g of Y in terms of F, f and ψ .
- (b) (9 points) Let $X \sim \mathcal{N}(0, 1)$. Let $Y = X^4$. Calculate the p.d.f. of Y .
- T. 3.** (a) (3 points) Define the moment generating function $t \mapsto M(t)$ of the random variable X .
- (b) (2+5 points) Write down the formula that connects the first/second derivative of M and the first/second moment of X . Prove the formula relating the first derivative of M and the expectation of X in the case when X is absolutely continuous and f denotes its p.d.f.
- (c) (7 points) Derive the formula for the moment generating function M of the standard normal distribution and calculate the variance of the standard normal distribution using M .

Formula sheet: $X \sim \text{EXP}(\lambda): \mathbb{E}(X) = \frac{1}{\lambda}, \text{Var}(X) = \frac{1}{\lambda^2}$. $X \sim \text{POI}(\lambda): \mathbb{E}(X) = \text{Var}(X) = \lambda$.

- P. 1.** (17 points) The numbers $1, 2, \dots, 6$ are distributed randomly (without repetition) among six players, A, B, C, D, E and F. First A and B compete: whoever has the higher number wins the round. The one that wins the first round will now face C in the second round, then the one that wins this round will play against D, then the winner of this round competes with E, then with the winner of this round competes with F. Let X denote the number of rounds that A wins. Find $\mathbb{E}(X)$.
- P. 2.** (16 points) A factory produces two types of coins: a fair one and a rigged one, which has a probability of 56% showing heads. We have such a coin, but we don't know if it is fair or rigged. To decide about this, we perform the following statistical test: we flip the coin 1000 times, and if it shows heads at least 530 times, we declare it to be rigged, but if there are less than 530 heads among the coin tosses then we declare the coin to be fair. What is the probability that our test is wrong if the coin is rigged?
- P. 3.** (8+9 points) In the city of Quahog, the fire department receives phone calls for two possible reasons: false fire alarms by pranksters and actual fires. These phone calls arrive according to independent Poisson point processes. On average, there are two actual fires per week. On average, 12 hours pass between two consecutive prank phone calls. Let X denote the number of prank calls before the first actual fire, starting from now. Compute $\mathbb{E}(X)$ and $\text{Var}(X)$. *Hint: First compute $\mathbb{E}(X)$ and $\mathbb{E}(X^2)$ using the tower rule of conditional expectation by conditioning on the time of the first real fire.*

Bonus: (10 points) What is the distribution of X ?

NAME: NEPTUN-CODE:

Probability theory 1 exam, 2024. Jan. 16.

*Working time: 100 min. Only simple, non-programmable calculators are allowed,
standard normal distribution table on the other side.*

The achievable maximum score (with the Bonus exercise) is 110 points, but we consider 100 points as 100%.

- T. 1.** (a) (5 points) State and prove the law of total probability! (b) (6 points) State and prove Bayes' theorem.
(c) (5 points) Assume that. $E_4 \subseteq E_3 \subseteq E_2 \subseteq E_1$. Write down and prove the relation that expresses $\mathbb{P}(E_4)$ in terms of $\mathbb{P}(E_1)$, $\mathbb{P}(E_2 | E_1)$, $\mathbb{P}(E_3 | E_2)$, and $\mathbb{P}(E_4 | E_3)$.
- T. 2.** (a) (2+2+5 points) Define the $\text{GEO}(p)$ distribution based on its intuitive meaning, and use it to calculate its weight function. Calculate the moment generating function of $\text{GEO}(p)$.
(b) (2+3 points) Define the negative binomial distribution based on its intuitive meaning and use it to calculate its weight function.
(c) (3 points) Calculate the moment generating function of the negative binomial distribution. *Hint:* What is the moment generating function of the sum of independent random variables?
- T. 3.** (a) (4 points) Let the pair of variables (X, Y) be jointly absolutely continuous, denote their joint density function by $f(x, y)$. Define the following notions (i.e., write a formula for them using f): the probability density function f_Y of the marginal distribution Y , the conditional density function $f_{X|Y}$ of X given $Y = y$, the conditional expectation of X given $Y = y$.
(b) (13 points) The signal X emitted by a radio antenna is a random variable with $\mathcal{N}(\mu, 1)$ distribution. The received signal Y is obtained by adding a noise Z (independent from X) to the signal X , where Z has $\mathcal{N}(0, 1)$ distribution. Determine the conditional density function of X given $Y = y$, moreover calculate the conditional expectation of X given $Y = y$.
- P. 1.** (17 points) Let $f : [0, 1] \rightarrow [0, 1]$ be a continuous function, and let $I = \int_0^1 f(x)dx$. Let $U_1, U_2, \dots, U_n$ and $V_1, V_2, \dots, V_n$ be i.i.d. with uniform distribution on $[0, 1]$. Let X_n denote the number of indices $i \in \{1, \dots, n\}$ for which $V_i < f(U_i)$. We estimate I by X_n/n . How big should we choose n if we want the error of our estimation to be greater than 0.05 with probability at most 0.01, if
(a) (10 points) $f(x) = x^4$?
(b) (7 points) we apply this method to an unknown continuous function $f : [0, 1] \rightarrow [0, 1]$?
- P. 2.** (16 points) The chroniclers of ancient Egypt distinguished two types of years: scarce years and abundant years. Successive years are scarce or abundant, independently from each other. On the long run, half of the years are scarce and half are abundant. Observing the Egyptian chronicle for n years, let us denote by X_n the number of scarce years that are followed by a scarce year and let us denote by Y_n the number of abundant years that are followed by an abundant year. Calculate the limit of the correlation coefficient of X_n and Y_n as $n \rightarrow \infty$!
- P. 3.** Let X and Y have bivariate normal distribution. Assume that $\mathbb{E}(X) = 1$, $\mathbb{E}(Y) = 0$, $\text{Var}(X) = 1$, $\text{Var}(Y) = 4$ and $\text{Cov}(X, Y) = 0$. Below, M_i , $i = 1, 2, 3, 4$ denote planar domains, and the task is to calculate $\mathbb{P}((X, Y) \in M_i)$. *Hint:* centralization (since $\mathbb{E}(X) \neq 0$) + normalization = standardization, and then draw M_i !
(a) (5 points) $M_1 = \{(x, y) : |x| \leq 1, |y| \geq 1\}$
(b) (6 points) $M_2 = \{(x, y) : 0 \leq x \leq 2, y \leq x - 1\}$
(c) (6 points) $M_3 = \{(x, y) : 4x^2 - 8x + y^2 \leq 0\}$
- Bonus:** (10 points) $M_4 = \{(x, y) : x \leq 1, y \geq 0, y \leq 2x\}$

NAME: NEPTUN-CODE:

Probability theory 1 exam, 2024. Jan. 23.

*Working time: 100 min. Only simple, non-programmable calculators are allowed,
standard normal distribution table on the other side.*

The achievable maximum score (with the Bonus exercise) is 110 points, but we consider 100 points as 100%.

- T. 1.** (a) (2+2+3+3 points) Write down the probability weight function of the *binomial* and *Poisson* distributions and calculate (from the weight function or using another learned method) the expected value of both distributions.
- (b) (7 points) In what limiting case and exactly in what sense can the binomial distribution be approximated by the Poisson distribution? State and prove the theorem you have learned about this.
- T. 2.** (a) (6 points) State and prove the Cauchy-Schwarz inequality for the random variables X and Y .
- (b) (6 points) How do we define the correlation coefficient $\rho(X, Y)$ of X and Y ? What is the maximum possible value of it? In what case do we get this extreme value for $\rho(X, Y)$? Prove your claims.
- (c) (6 points) State and prove Steiner's theorem!
- T. 3.** Let X and Y be independent random variables with probability density functions $f_X(x)$ and $f_Y(y)$, and let $Z = X + Y$.
- (a) (8 points) What formula gives the cumulative distribution function and probability density function of Z ? Prove your claims.
- (b) (2 points) How do we define the moment generating function of X ?
- (c) (5 points) How do we get the moment generating function of Z from the moment generating functions of X and Y ? Prove your claim.
- P. 1.** 60% of the border guards of Ruritania are superficial, while 40% of them are thorough. Superficial border guards stop cars that try to enter the county with probability 0.6, thorough ones stop cars with probability 0.9, independently of each other. I want to drive into Ruritania, and I see a blue car and a red car in front of me. The border guard stopped the blue car, but did not stop the red car.
- (a) (9 points) What is the probability that this border guard is superficial in nature?
- (b) (8 points) Knowing the fate of the previous two cars, what is the probability that this border guard will stop me?
- P. 2.** (16 points) The lifetime of the „Mehr Licht” type light bulb is exponentially distributed. According to the manufacturer's measurements, 80 percent of the light bulbs survive for at least one year. How should the manufacturer set the expiration date (i.e., the time limit of guaranteed performance) of the light bulb if they want no more than 2 percent of customers to complain?
- Bonus:** (10 points) I use this type of light bulb in my study and replace it immediately when it burns out. Determine the distribution function for the time of the fourth light bulb change (if year is the unit of measurement of time)! *Hint:* What do the lightbulb change times form?
- P. 3.** Let $X_1, X_2, \dots, X_n, X_{n+1}$ be independent random variables with $\mathcal{N}(0, 1)$ distribution, and define the random variable Y with the formula $Y = \frac{X_1 + \dots + X_n}{n}$.
- (a) (6 points) Determine the covariance matrix of the random two-dimensional vector (X_1, Y) .
- (b) (11 points) $\mathbb{P}(|Y| \leq |X_{n+1}|) = ?$

NAME: NEPTUN-CODE:

Probability theory 1 exam, 2024. Feb. 9.

*Working time: 100 min. Only simple, non-programmable calculators are allowed,
standard normal distribution table on the other side.*

The achievable maximum score (with the Bonus exercise) is 110 points, but we consider 100 points as 100%.

- T. 1.** (a) (1+2+4 points) State the inclusion-exclusion formula for two events, three events, and n events.
(b) (10 points) A group of n men went to have dinner in a restaurant. They left their hats in the wardrobe. After having dinner and drinking some wine, they took their hats out of the wardrobe completely at random. What the probability that at least one member of the group wore his own hat on the way home? Calculate the limit of this probability in the $n \rightarrow \infty$ limit.
- T. 2.** (a) (1+1+3+3 points) Define the geometric distribution based on its intuitive meaning. Calculate its probability mass function, its expected value, and its variance.
(b) (2+3+4 points) Define the hypergeometric distribution based on its intuitive meaning. Calculate its probability mass function and its expected value.
- Bonus:** (10 points) Calculate the variance of the hypergeometric distribution.
- T. 3.** (5+5+6 points) State and prove Markov's inequality, Chebyshev's inequality and the weak law of large numbers.
- P. 1.** The expected number of car accidents per day in the 11th district of Budapest is ten. The probability that the drivers involved in the accident will be able to agree on who is responsible is 0.6. If the drivers cannot agree, they call the police for an on-site inspection.
(a) (4 points) What distribution should we use to model the number of car accidents per day? Why?
(b) (6 points) What is the probability that the police will be called for at least three car accidents tomorrow?
(c) (7 points) Yesterday the police were called to the scenes of five car accidents. What is the probability that altogether ten car accidents occurred in the district yesterday?
- P. 2.** (4+3+5+5 points) The joint density function of X and Y is $f(x, y) = \frac{1}{y} e^{-(y+x/y)} \mathbb{1}[x > 0, y > 0]$.
(a) Determine the density function of Y . (b) Determine the expected value of Y . (c) Determine the conditional density function of X under the condition $Y = y$. (d) Determine the expected value of X .
- P. 3.** Mr. X travels to work by train and long-distance bus. On the way to work, he misses the connection with a probability of $1/5$, and on the way home with he misses the connection with probability of $1/4$. Let us assume that there are 220 working days per year, what is the approximate probability that Mr. X
(a) (6 points) will miss the connection at least forty times this year on his way to work?
(b) (10 points) will miss the connection more times on the way home than he will on the way to work next year?

Name: NEPTUN code: Major:

Lecturer: Balázs Bárány

Probability Theory 1 exam, December 10th, 2024

Working hours: 100 min. Non-programable calculator without internet connection can be used.

Maximal amount of points (with the Bonus): 110 points, but 100 points are already considered as 100%.

- The. 1.** (a) (3 points) Define the conditional probability of the event A conditioned on event B .
(b) (5 points) Let $E_1, \dots, E_n$ be arbitrary events. Give and prove the relation between the probability $\mathbb{P}(\cap_{i=1}^n E_i)$ and the probabilities $\mathbb{P}(E_1), \mathbb{P}(E_2 | E_1), \dots, \mathbb{P}(E_n | E_1 \cap \dots \cap E_{n-1})$!
(c) (3 points) When do we call a countable collection of events partition (complete event system)?
(d) (5 points) State and prove the theorem of complete event systems (also known as *law of total probability*)!
- The. 2.** (a) (5 points) State and prove Markov's inequality.
(b) (5 points) State and prove Chebisev's inequality.
(c) (7 points) State and prove the Weak Law of Large Numbers.
- The. 3.** (a) (3 points) Define the Poisson distribution,
(b) (3 points) Prove that it is indeed a distribution, that is, if X has Poisson distribution then $\sum_{k=0}^{\infty} \mathbb{P}(X = k) = 1$,
(c) (3 points) Compute the moment generating function of the Poisson distribution,
(d) (4 points) Compute the expected value of the random variable X with Poisson distribution,
(e) (4 points) Compute the variance of X .
- Prac. 1.** The Advent calendar is a 24-day calendar from the 1st to the 24th of December in which a small candy is hidden for each day: kids open one door of their calendar each morning. The little angels put four types of chocolate in our Advent calendar: white, milk, dark and nougat, six of each.
(a) (8 points) Find the probability that until December 10th (included) we have found at least one of all four types of chocolate.
(b) (9 points) Determine the covariance of the number of dark and nougat chocolates we found until 10th of December (included).
- Bonus:** (10 points) The Santa Claus got a little bit dizzy and he visited us not on Christmas but on a random day uniformly chosen between 1st and 24th of December. Find the expected value and the variance of the number of milk chocolates we found until the day of Santa Claus' visit. *Hint 1:* If we consider a random variable X with hypergeometric distribution, where N is the population size, K is the number of success states in the population, n is the sample size and X is the number of success states in the sample then $\text{Var}(X) = n \frac{K}{N} \frac{N-K}{N} \frac{N-n}{N-1}$. *Hint 2:* If $a < b$ are positive integers and Y is uniformly distributed on the set $\{a, a+1, \dots, b-1, b\}$ then $\text{Var}(Y) = \frac{(b-a+1)^2-1}{12}$.
- Prac. 2.** (16 points) Choose a point uniformly and randomly in the interior of an equilateral triangle of unit side length. Denote by D the distance of the point from the closest side of the triangle. Determine the distribution function F and the density function f of D . Plot the graph of the distribution function $F : \mathbb{R} \rightarrow [0, 1]$.
- Prac. 3.** On the exam of a subject, the expected value of a student's score is 74 points, while the standard deviation is 14. Assume that the scores of students are independent of each other. The first exam was taken by 36 students and the second was taken by 49. Estimate the following probabilities:
(a) (9 points) the average score on the two exams differ by more than 2 points,
(b) (8 points) the average score of the second exam is closer to 74 than that of the first exam's average score.

Name: NEPTUN code: Major:

Probability Theory 1 exam, December 17th, 2024

Working hours: 100 min. Non-programable calculator without internet connection can be used.

Maximal amount of points (with the Bonus): 110 points, but 100 points are already considered as 100%.

- The. 1.** (a) (3 points) State the axioms of a probability measure.
(b) (4 points) State the inclusion-exclusion (sieve) formula for n events.
(c) (3 points) Prove the inclusion-exclusion (sieve) formula for 2 events.
(d) (6 points) The 5th class of a primary school with 29 students is preparing for Christmas. They decided that they will give a small gift for each other. To decide who gives a gift to whom, they write their names on a little slip of a paper and they put it in a hat. They shuffle the slips in the hat and everybody draws one name randomly. What is the probability that no student will draw his/her own name? *Instruction:* The final result is not enough, present the computation!
- The. 2.** Let $\mu \in \mathbb{R}$ and $\sigma \in \mathbb{R}$ with $\sigma \neq 0$.
(a) (3 points) Write down the probability density function (p.d.f.) of the $N(\mu, \sigma^2)$ distribution.
(b) (7 points) Prove that the p.d.f. of the standard normal distribution $N(0, 1)$ is indeed a p.d.f.
(c) (6 points) Prove that if Y has distribution $N(0, 1)$ then $X = \sigma \cdot Y + \mu$ has distribution $N(\mu, \sigma^2)$.
- The. 3.** (a) (2+2 points) Define the covariance and the correlation of two random variables X and Y .
(b) (2 points) Define the covariance matrix of the random variables $X_1, \dots, X_n$.
(c) (5 points) Show that the covariance matrix is positive semi-definite.
(d) (6 points) State and prove the Cauchy-Schwartz inequality.
- Bonus:** (10 points) Let X be a random variable uniformly distributed on the interval $[6, 9]$. The distribution of the random variable Y is unknown but we know that its expected value is 10, its deviation is $\sqrt{3}$ and $\text{Cov}(X, Y) = 3/2$. Using these informations, what can we say about the probability $\mathbb{P}(Y \geq 9)$?
- Prac. 1.** (8+8 points) We put five butterfly traps on a field far from each other. Every trap, independently of the others, opens with probability $2/3$. If a trap does not open it won't catch any butterfly. If a trap is open then every butterfly nearby flies into the trap with a small probability independently of the other butterflies. For an opened trap, the expected time between trapping two butterflies is 20 minutes. The traps are on the field for one hour and then we collect them. What is the expected value and variance of the caught butterflies?
- Prac. 2.** (17 points) Let the side-lengths of a rectangle be 4 and 3. Let us choose two points, one-one on both sides with length 4, uniformly, randomly and independently. Denote by X the distance of these points. Find the distribution function and the density function of X . Plot the graph of the distribution function.
- Prac. 3.** Santa Claus drives his reindeer-powered flying sleigh, while baby Jesus drops gifts from the back seat through the chimneys of family homes. Every time he targets the middle of the chimney-holes (of which we assume that it is the origin of a planar coordinate system), but the point where the gift lands is a two-dimensional jointly normal random variable with expected value $(0, 0)$ and covariance matrix 3-times the identity matrix. What is the probability that he hits the chimney-hole if
(a) (8 points) the chimney-hole is square shaped with vertices $(0, 1)$, $(1, 0)$, $(0, -1)$ and $(-1, 0)$?
(b) (9 points) the chimney-hole is disc shaped centered at the origin and with radius $3/2$?

Name: NEPTUN code: Major:

Probability Theory 1 exam, January 7th, 2025

Working hours: 100 min. Non-programable calculator without internet connection can be used.

Maximal amount of points (with the Bonus): 110 points, but 100 points are already considered as 100%.

- The. 1.** (a) (3 points) Define when the events A and B are independent.
(b) (4 points) Prove that if the events A and B are independent then A and B^c are independent too (where B^c denotes the complement of B).
(c) (3 points) Define when the events A and B are conditionally independent with respect to the event C .
(d) (6 points) Prove that if the events A and B are conditionally independent with respect to the event C then $\mathbb{P}(A|B \cap C) = \mathbb{P}(A|C)$.
- The. 2.** (8+8 points) Describe Bertrand's paradox! That is, show at least two different methods for choosing a "uniformly distributed" random chord of a circle, and calculate in these cases the probability that the length of the chord is longer than the side-length of the equilateral triangle inscribed in the circle.
Bonus: (3 points) If you write down all three different methods learnt in class.
- The. 3.** (a) (2+2 points) Define the covariance and the correlation of two random variables X and Y .
(b) (1+4 points) Define the covariance matrix and show that it is positive semi-definite.
(c) (4 points) State and prove the Cauchy-Schwartz inequality.
(d) (5 points) Show that if the correlation of X and Y is 1 then $Y = a \cdot X + b$ for some $a \in \mathbb{R}_+$ and $b \in \mathbb{R}$.
- Prac. 1.** (8+8 points) We have two urns with red and blue balls. In the first urn, there are 4 red and 5 blue balls, while in the second urn, there are 3 red and 7 blue balls.
(a) Scenario 1: we take a ball from the first urn and put it into the second urn. We then take a ball from the second urn and put it into the first urn. Then, we take another ball from the first urn. What is the probability that this ball is red?
(b) Scenario 2: we take 1-1 ball from the two urns at the same time, and switch them. Then we take a ball from the first urn. What is the probability that this ball is red?
- Prac. 2.** The ambulance and fire departments are not assigned to the same areas. Kukutyin (800 inhabitants) and Piripócs (1200 inhabitants) are part of the ambulance rescue area "KP", while Szecső (900 inhabitants) and Pajkaszeg (1300 inhabitants) are part of the ambulance rescue area "SP". Kukutyin and Szecső are in the fire alarm area "KS", and Piripócs and Pajkaszeg are in the fire alarm area "PP". On New Year's Eve, each resident, independent of the rest of the population, has a 0.001 chance of being a victim of a fireworks accident. The ambulance and the fire-fighters are both alarmed in case of each fireworks accident. *Instructions:* Use the Poisson approximation to solve the following problems.
(a) (5 points) What is the probability that the fire department in the fire alarm area "PP" will receive at least two alarms?
(b) (6 points) Assuming that the ambulance in the ambulance rescue area "KP" received exactly three alarms, what is the probability that there were exactly two firework accidents in Piripócs?
(c) (6 points) Denote by X the number of alarms received by the fire department in the fire alarm area "PP", and denote Y the number of alarms received by the ambulance in the rescue area "KP". $\text{Cov}(X, Y) = ?$
Bonus: (4+3 points) Calculate $\mathbb{E}(X|Y)$ and $\mathbb{D}^2(X|Y)$.
- Prac. 3.** (17 points) Two groups of researchers want to measure the charge of a particle. The first group takes 5 measurements and averages it, the second group takes 7 measurements and also averages it. The measurements are independent and identically distributed with normal distribution, the expected value is the actual charge μ of the particle, and the standard deviation is σ . What is the probability that the first group's result is more accurate than the second group's result?

NAME: NEPTUN CODE: MAJOR:

Probability Theory Exam, January 14, 2025

Working time: 100 min. Only simple, non-programmable calculators are allowed, standard normal distribution table on the other side. The achievable maximum score (with the Bonus exercise) is 110 points, but we consider 100 points as 100%.

- T. 1.** Let (X, Y) be jointly absolutely continuous random variables with joint density function $f(x, y)$.
- (a) (1+1+1 points) Define the marginal density function $f_X(x)$, the conditional density function $f_{Y|X}(y|x)$, and the conditional expectation $\mathbb{E}(Y|X = x)$ using formulas.
 - (b) (1+4 points) State and prove the tower property of conditional expectation.
 - (c) (2 points) State the formula for the conditional variance ($\text{Var}(Y) = \dots$).
 - (d) (3+4 points) Let $X \sim \text{EXP}(\lambda)$ and given a fixed $X = x$, let Y have the distribution $\text{UNI}[0, x]$. Compute the expected value and variance of Y . *Hint: The variance of $\text{UNI}[0, x]$ is $x^2/12$.*
- T. 2.** (a) (3+2+2 points) Let $(X_1, \dots, X_n)$ be a general n -dimensional jointly normal random vector. Write down the joint density function of $(X_1, \dots, X_n)$, explaining the dimensions of the vectors or matrices in the formula, the covariance matrix, and the relationship between the matrix in the density function and the covariance matrix.
- (b) (5 points) Let (X, Y) be jointly normal with $\text{Cov}(X, Y) = 0$. Prove that X and Y are independent.
 - (c) (5 points) Give an example of a pair of random variables (X, Y) that are not independent but satisfy $\text{Cov}(X, Y) = 0$.
- T. 3.** (a) (2+1+4 points) Define the (optimistic) geometric distribution with parameter p (both with a formula and an intuitive explanation) and compute its expected value.
- (b) (2+2 points) Define the memoryless property and prove that the geometric distribution has this property.
 - (c) (5 points) Let X and Y be independent random variables with $\text{GEO}(p)$ distribution. Compute the conditional probability $\mathbb{P}(X = 6 | X + Y = 15)$.
- P. 1.** After the holiday sales, Thomas bought 1000 light bulbs because they were on sale. The lifetime of each bulb follows an exponential distribution with an expected value of 3 months. If a bulb burns out, Thomas (or his descendants) immediately replace it with a new one.
- (a) (3 points) What is the expected number and variance of bulb replacements in the first year?
 - (b) (5 points) Compute the distribution function of the time of the second bulb replacement. *Hint: The time of the second bulb replacement is greater than t if and only if there are fewer than two replacements in the interval $[0, t]$.*
 - (c) (9 points) Approximate the largest T such that the bulb supply lasts for T years with a 90% probability. (A standard normal distribution table is on the back.)
- P. 2.** Determine the density functions of X and Y if
- (a) (8 points) ξ is uniformly distributed on the interval $[-3, 2]$, and $X = \xi^2$,
 - (b) (8 points) ϑ follows an $\text{EXP}(\lambda)$ distribution, and $Y = \sqrt{\vartheta}$.
- Bonus (10 points)** Let $X \sim \text{EXP}(\lambda_1)$ and $Y \sim \text{EXP}(\lambda_2)$ be two independent random variables. Compute the density function of $Z = X + Y$.
- P. 3.** Let X and Y be independent normally distributed random variables. X has mean 0 and variance 4, and Y has mean 0 and variance 9. Compute the probability $\mathbb{P}((X, Y) \in H)$ where
- (a) (8 points) $H = \{(x, y) \in \mathbb{R}^2 : x \leq -y, -1 \leq y \leq 1, x \geq -3\}$,
 - (b) (9 points) $H = \{(x, y) \in \mathbb{R}^2 : (\frac{x}{4})^2 + (\frac{y}{9})^2 \leq 2\}$.

Name: NEPTUN code: Major:

Probability Theory 1 exam, January 23rd, 2025

Working hours: 100 min. Non-programable calculator without internet connection can be used.

Maximal amount of points (with the Bonus): 110 points, but 100 points are already considered as 100%.

- The. 1.** (a) (2+2 points) Define the binomial distribution $\text{BIN}(n, p)$ based on its descriptive meaning and derive from this the probabilities $\mathbb{P}(X = k)$ for the possible values of k if $X \sim \text{BIN}(n, p)$.
(b) (3+4 points) Determine the expected value and variance of $\text{BIN}(n, p)$.
(c) (2+4 points) State and prove the Poisson approximation theorem.

Bonus (10 points) Let $X \sim \text{BIN}(n, p)$. Give a closed formula for $\mathbb{E}(\frac{1}{X+1})$.

- The. 2.** (a) (6 points) Define the probability density function $f_X(x)$ of an absolutely continuous random variable X . Give and prove the defining properties of f_X .
(b) (10 points) Let Y be a non-negative absolutely continuous random variable. Show that $\mathbb{E}(Y) = \int_0^\infty \mathbb{P}(Y > y) dy$.

- The. 3.** Let (X, Y) be jointly absolutely continuous random variables with joint density function $f(x, y)$.
(a) (1+1+1 points) Define the marginal density f_X , the conditional density $f_{Y|X}(y|x)$ and the conditional expectation $\mathbb{E}(Y|X)$.
(b) (1 points) State the tower rule.
(c) (1 points) Using the conditional expectation, define the conditional variance $\mathbb{D}^2(X|Y)$.
(d) (1+5 points) State and prove the formula for conditional variance.
(e) (2+4 points) Denote the number of customers of a shop on a day by N , which is random. Let ξ_i be the spending of the i th customer, which is also random. We know that $\mathbb{E}(N) = 200$, $\mathbb{D}^2(N) = 81$, moreover, $\mathbb{E}(\xi_i) = 5000$ and $\mathbb{D}^2(\xi_i) = 10000$ for every i . Suppose that $N, \xi_1, \xi_2, \dots$ are independent. Denote Y the total income of the shop on a day. Find the expected value and variance of Y .

Prac. 1. In the Bonferroni factory, there are two production lines: every third car is made on the first line, while the rest is made on the second. Every Bonferroni car stalls at the ignition with a small probability (independently at each ignition). However, this probability depends on the production line where the car was made. The probability that a car made on the first production line stalls at least once in a year is $1/4$, while the probability of the same is $1/2$ on the second production line.

- (a) (8 points) Lewis is an average customer who bought a Bonfferroni car and he wants to use it for two years. What is the probability that Lewis' car stalls exactly two times during this two years?
(b) (8 points) Max is an another average customer who is already using his Bonfferroni car for a year, and during this year his car did not stall at all. What is the probability that Max's car won't stall in the next year?

Prac. 2. (8+9 points) We throw an unbiased dice 30 times. Let X be the number of occasions when an even number is followed by an odd. What is the expected value and variance of X ?

Prac. 3. (17 points) Given 48 random real numbers, we round them one by one to the closest integer. Assume that the rounding errors for different numbers are independent, identically distributed, and uniform on the interval $[-\frac{1}{2}, \frac{1}{2}]$. Denote S the sum of these 48 numbers and denote T the sum of the integers that we obtain after rounding each of the summands. What is the probability that we get T by rounding S ? Use normal approximation. (Hint: if $\xi \sim \text{UNI}(a, b)$ then $\mathbb{D}^2(\xi) = \frac{(b-a)^2}{12}$.)

NAME: NEPTUN-CODE: MAT or PHYS?:

Probability theory exam, 2025. dec. 18.

*Working time: 100 minutes. Non-programmable, offline calculator can be used,
standard normal distribution table on the other side.*

Maximum possible score (including bonus): 110 points, but even 100 points is considered as 100%.

- The. 1.** (a) (4 points) Define the expected value and standard deviation of discrete probability distributions.
(b) (4 points) Give an example of a distribution with a finite expected value but an infinite variance.
(c) (4 points) There are two formulas for the standard deviation (one: the squared distance from the expected value; the other: expressed in terms of the first and second moments). Write down the formula that you did not write down in task (a) and prove that the two formulas give the same value.
(d) (4 points) Derive the formula for the standard deviation of the binomial distribution.
- The. 2.** Let X and Y be independent absolutely continuous random variables. Let $Z = X + Y$. Derive the formula for the distribution function of Z (6 points). From this, derive the formula for the density function of Z (3 points). If X and Y are both uniformly distributed on the interval $[2, 4]$, then give the formula for the density function of Z and draw the graph of the density function of Z (4 points), and calculate the probability $\mathbb{P}(Z > 7)$ (4 points).
- The. 3.** Let (X, Y) be jointly absolutely continuous. Define the conditional expectation of X under the condition $Y = y$ (3 points), and the conditional expectation of $\mathbb{E}(X | Y)$ (2 points). State and prove the tower rule of conditional expectation (5 points). Define the meaning of the terms in the conditional variance formula (3 points) and prove the formula (4 points).
- Pr1. 1.** Gaspard carries 3 pices of gold, 2 frankincense and 1 myrrh in his gift package, Melchior carries 2 pieces of gold, 1 frankincense and 3 myrrh in his package, and Balthazar carries 1 piece of gold, 3 frankincense and 2 myrrh in his package. The three Wise Men arrived at the Holy Family in random order. Mary blindly reaches into the gift package of the wise man who arrived first and takes out two gifts.
(a) (6 points) What is the expected number of myrrhs in Mary's hand? *Hint:* It's easier if you use the method of decomposition as a sum of indicators!
(b) (6 points) Assuming that Mary holds a piece of gold and a piece of frankincense, what is the probability that Gaspar arrived first?
(c) (5 points) Assuming that she holds a piece gold and a piece of frankincense, what is the expected number of myrrhs in the same package?
- Pr2. 2.** (16 points) We roll two independent fair dice. What is the joint probability weight function of X and Y if X is the minimum of the two throws and Y is their sum?
- Pr3. 3.** On average, the Tume parcel delivery company receives an online Christmas gift order every two seconds. Independently of this, the rival Azla parcel delivery company also receives an order every two seconds on average.
(a) (4 points) What is the variance of the time of the fifth order received by Azla after 10.00am?
(b) (5 points) What is the variance of the number of orders received by both companies together in the next 10 seconds?
(c) (8 points) Estimate the probability that Tume will receive at least 50 more orders than Azla in the next hour. *Instructions:* Φ table on the opposite page.
- Bonus** (10 points) Estimate the probability of the joint occurrence of events that Azla will receive more orders between noon and 6 pm today and also between noon and midnight today. *Instructions:* Φ table on the opposite page.

Name: NEPTUN code: Major:

Probability Theory 1 exam, 6th January, 2026

Working hours: 100 min. A non-programmable calculator without internet connection can be used.

Maximal amount of points (with the Bonus): 110 points, but 100 points are already considered as 100%.

- The. 1.** (a) (2+2 points) Define the binomial distribution $\text{BIN}(n, p)$ based on its descriptive meaning and derive from this the probabilities $\mathbb{P}(X = k)$ for the possible values of k if $X \sim \text{BIN}(n, p)$.
(b) (3+4 points) Determine the expected value and variance of $\text{BIN}(n, p)$.
(c) (2+6 points) State and prove the Poisson approximation theorem.

- The. 2.** (a) (2 points) Define the cumulative distribution function (c.d.f.) $F_X(x)$ of a random variable X .
(b) (4+8 points) State and prove the defining properties of the c.d.f.
(c) (2 points) Define the probability density function (p.d.f.) $f_X(x)$ of an absolutely continuous random variable X .
(d) (3 points) State the relation between the c.d.f. and the p.d.f. of an absolutely continuous random variable.

The. 3. Let $(\Omega, \mathcal{F}, \mathbb{P})$ be a probability space.

- (a) (2+2 points) Define when we call two events $A, B \in \mathcal{F}$ independent. Define when we call a countable collection of events $A_1, A_2, \dots$ independent.
(b) (2 points) Define when we call two random variables $X: \Omega \rightarrow \mathbb{R}$ and $Y: \Omega \rightarrow \mathbb{R}$ independent.
(c) (3 points) Give equivalent conditions for two random variables being independent (in the discrete case and the jointly absolutely continuous case)
(d) (1+2 points) Define the covariance $\text{cov}(X, Y)$ of X and Y and calculate the covariance of a pair of independent, jointly absolutely continuous random variables X and Y .

Prac. 1. An archer is practicing shooting. The bull's eye (i.e. the center of his target) is the origin of the coordinate system, and the impact coordinates (X, Y) of his arrows are random variables such that $X \sim N(0, \frac{\sigma^2}{100})$, $Y \sim N(0, \frac{\sigma^2}{100})$, and X and Y are independent, where σ is the distance of the archer from the target in meters.

- (a) (4 points) What is the probability that the archer hits the rectangle $[-2, 2] \times [-1, 1]$ if he stands exactly $\sigma = 10$ meters away from the target?
(b) (7 points) What is the probability that the archer hits the square $\{(x, y) : |x| + |y| \leq 2\}$ if he stands exactly $\sigma = 20$ meters away from the target?
(c) (8 points) If the target he aims is a disc of radius 1 meter (centered at the origin), what is the probability that the archer hits it as a function of σ ?

Bonus: (10 points) In the previous exercise, derive the explicit formula for the expected Euclidean distance of the impact point of the arrow from the bull's eye as a function of σ .

Prac. 2. Let $X_1, X_2, \dots$ and $Y_1, Y_2, \dots$ be independent and identically distributed random variables with distribution $\text{EXP}(\lambda)$. Let $P_n = (\sum_{k=1}^n X_k, \sum_{k=1}^n Y_k)$ for $n = 1, 2, \dots$ and let $\mathcal{S}$ be the following random set of random points on the plane: $\mathcal{S} = \{P_n : n = 1, 2, \dots\}$. In other words: $\mathcal{S}$ consists of $(X_1, Y_1), (X_1 + X_2, Y_1 + Y_2), \dots$

- (a) (4 points) What is the probability that there are no points from $\mathcal{S}$ in the square $[0, t] \times [0, t]$?
(b) (6 points) What is the probab. that there are at least two points from $\mathcal{S}$ in the square $[0, t] \times [0, t]$?

Prac. 3. Santa's elves are in a hurry. There are 5 socks and 10 elves. Each elf throws one gift in a randomly chosen sock.

- (a) (6 points) What is the expected value of the number of gifts in the first sock?
(b) (7 points) What is the expected value of the number of non-empty socks?
(c) (8 points) What is the variance of the number of non-empty socks?

Name: NEPTUN code: Major:

Probability Theory 1 exam, 13th January 2026

Working hours: 100 min. A non-programmable calculator without an internet connection can be used.

Maximal amount of points (with the Bonus): 110 points, but 100 points are already considered as 100%.

The. 1. Let $(\Omega, \mathcal{F}, \mathbb{P})$ be a probability space.

- (a) (2 points) Define the conditional probability with respect to an event E .
- (b) (2 points) Give the definition of complete event systems (partitions)!
- (c) (2+3 points) State and prove the theorem of complete event systems (also known as the law of total probability).
- (d) (2+3 points) State and prove Bayes' Theorem.

The. 2. (a) (2 points) Give the probability density function (p.d.f.) of a random variable with normal distribution $N(\mu, \sigma^2)$.

- (b) (7 points) Prove that it is indeed a p.d.f. *Hint: why is it enough to prove it in the standardized case?*
- (c) (4 points) Give the joint probability density function of the random variables (X, Y) with joint normal distribution.
- (d) (7 points) Suppose that $X \sim N(\mu_1, \sigma_1^2)$ and $Y \sim N(\mu_2, \sigma_2^2)$ and X and Y are independent. Give the distribution of $X + Y$, and verify it. *Hint: why is it enough to prove it in the standardized case?*

The. 3. Let $(\Omega, \mathcal{F}, \mathbb{P})$ be a probability space, and let X and Y be random variables.

- (a) (2+2 points) Define the covariance $\text{cov}(X, Y)$ and the correlation $\text{corr}(X, Y)$ of X and Y .
- (b) (2+6 points) State and prove the Cauchy-Schwartz inequality.
- (c) (4 points) What can we say about the relation of X and Y if $\text{corr}(X, Y) = -1$? Verify your statement!

Prac. 1. The arrival times of the trams at the tram stops of Randomopolis follow a Poisson process. The Poisson processes of different tram lines are independent. On average, there are 12 trams per hour at a tram stop.

- (a) (5 points) I arrived to the stop at 8:00 a.m. What is the probability that I have to wait for a tram for more than 6 minutes?
- (b) (8 points) I will have an appointment at 8:40. To get there, I have to commute once. I take the first tram that arrives and travel with it for 15 minutes, when I get off. Then, at the stop where I got off, I waited for another tram and travelled with it for 10 minutes. What is the probability that I arrive at my appointment on time?
- (c) (5 points) On each of the trams that arrive at a particular stop, there are always 30 passengers, and each of them takes off independently with probability $1/3$. What is the expected value of the passengers taking off at the stop between 8:00 and 10:00?

Bonus: (10 points) What is the variance of the number of passengers taking off at the stop between 8:00 and 10:00?

Prac. 2. (16 points) I have two coins: an unbiased one and a biased one, which lands on heads with probability $3/5$. I forgot which is which. I picked one at random and tossed it 100 times; it landed heads 58 times. Estimate the probability that I picked the unbiased one using de Moivre-Laplace.

Prac. 3. Take a PQ line segment of length 2 and break it into two parts at a uniformly chosen point R . Let X denote the length of the line segment PR . We broke the other RQ segment again at a uniformly chosen point S . Let Y denote the length of the line segment RS .

- (a) (6 points) What is the joint probability distribution function of (X, Y) ?
- (b) (4 points) Find $\mathbb{E}(Y|X = x)$.
- (c) (6 points) Calculate $\text{cov}(X, Y)$.